

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon Web Services, VA -- station KHEF, America/New_York
Committed pair OSM 1000784035 -> 1059030596
Amazon Web Services / Amazon Web Services
Facade gap 69.7 m between the two halls
Weather record 42,554 real hourly records from KHEF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3448 C at 190 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-04-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	5.366	18.0	5.312	1.124
01	mechanical	yes	switch budget	5.242	18.0	4.101	1.314
02	mechanical	yes	switch budget	4.412	18.0	2.431	1.211
03	mechanical	yes	switch budget	4.692	18.0	2.991	1.169
04	mechanical	yes	switch budget	3.302	18.0	1.321	1.115
05	mechanical	yes	switch budget	1.937	18.0	0.211	1.081
06	mechanical	yes	switch budget	2.437	18.0	-0.399	1.387
07	mechanical	yes	switch budget	4.967	18.0	4.101	1.693
08	mechanical	yes	switch budget	8.857	18.0	9.101	2.422
09	mechanical	yes	switch budget	11.847	18.0	13.123	2.845

10	mechanical	yes	switch budget	16.357	18.0	16.881	2.978
11	mechanical	no	dry-bulb	19.283	18.0	20.110	2.294
12	mechanical	no	dry-bulb	21.637	18.0	22.991	1.808
13	mechanical	no	dry-bulb	23.409	18.0	24.208	1.474
14	mechanical	no	dry-bulb	24.819	18.0	24.783	1.360
15	mechanical	no	dry-bulb	26.198	18.0	25.573	1.266
16	mechanical	no	dry-bulb	25.680	18.0	25.255	1.333
17	mechanical	no	dry-bulb	25.352	18.0	24.764	1.462
18	mechanical	no	dry-bulb	24.716	18.0	23.119	1.761
19	mechanical	no	dry-bulb	21.907	18.0	18.541	1.924
20	mechanical	no	dry-bulb	20.247	18.0	15.771	2.037
21	FREE	yes	-	17.747	18.0	12.431	1.886
22	FREE	yes	-	15.242	18.0	11.321	1.767
23	FREE	yes	-	12.467	18.0	8.541	1.420

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.366 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.242 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.412 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.692 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.302 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.937 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.437 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.967 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.857 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.847 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.357 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.283 C against a 18.0 C limit, so it fails by 1.283 C. It would take a limit 1.283 C higher, or a bound 1.283 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.637 C against a 18.0 C limit, so it fails by 3.637 C. It would take a limit 3.637 C higher, or a bound 3.637 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.409 C against a 18.0 C limit, so it fails by 5.409 C. It would take a limit 5.409 C higher, or a bound 5.409 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.819 C against a 18.0 C limit, so it fails by 6.819 C. It would take a limit 6.819 C higher, or a bound 6.819 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.198 C against a 18.0 C limit, so it fails by 8.198 C. It would take a limit 8.198 C higher, or a bound 8.198 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.680 C against a 18.0 C limit, so it fails by 7.680 C. It would take a limit 7.680 C higher, or a bound 7.680 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.352 C against a 18.0 C limit, so it fails by 7.352 C. It would take a limit 7.352 C higher, or a bound 7.352 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.717 C against a 18.0 C limit, so it fails by 6.717 C. It would take a limit 6.717 C higher, or a bound 6.717 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.907 C against a 18.0 C limit, so it fails by 3.907 C. It would take a limit 3.907 C higher, or a bound 3.907 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.247 C against a 18.0 C limit, so it fails by 2.247 C. It would take a limit 2.247 C higher, or a bound 2.247 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.747 C, which is 0.253 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.886 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.3109 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.242 C, which is 2.758 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.767 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.3109 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.467 C, which is 5.533 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.420 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.3109 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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